DSA03-Natural Language Processing

List of Lab Program

26. Implement a machine translation program using the Hugging Face Transformers library, translate English text to French using python.

Code:

```
from transformers import MarianMTModel, MarianTokenizer

model_name = "Helsinki-NLP/opus-mt-en-fr"

tokenizer = MarianTokenizer.from_pretrained(model_name)

model = MarianMTModel.from_pretrained(model_name)

text = input("Enter English text: ")

# Tokenize and translate

tokenized_text = tokenizer(text, return_tensors="pt", padding=True)

translated_tokens = model.generate(**tokenized_text)

# Decode translation

translation = tokenizer.decode(translated_tokens[0], skip_special_tokens=True)

print("French Translation:", translation)
```

Output:

```
Enter English text: Hi,how are you?
French Translation: Bonjour, comment allez-vous ?
```